



Mismatch II

Mental health

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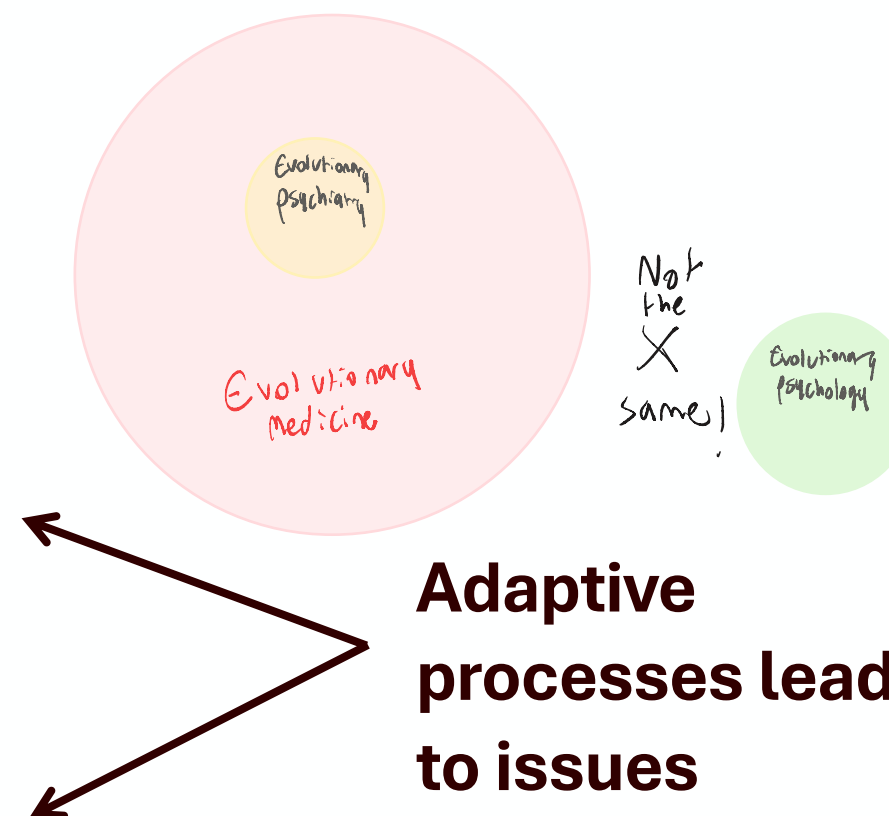
Mismatch: two types

Evolutionary mismatch

Mismatch between evolutionary environment and current environment

Developmental mismatch

Mismatch between developmental environment and current environment



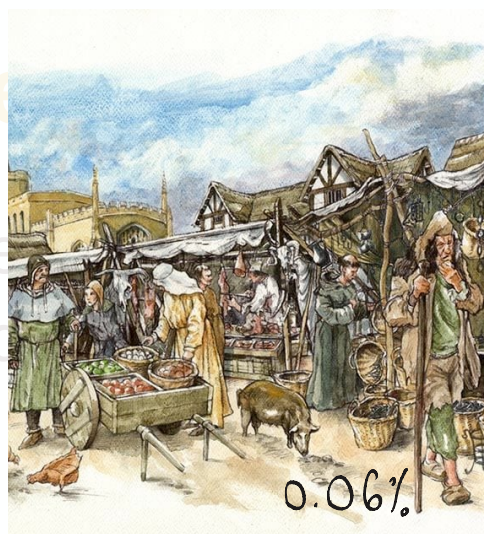
Mismatch: two types

Evolutionary mismatch

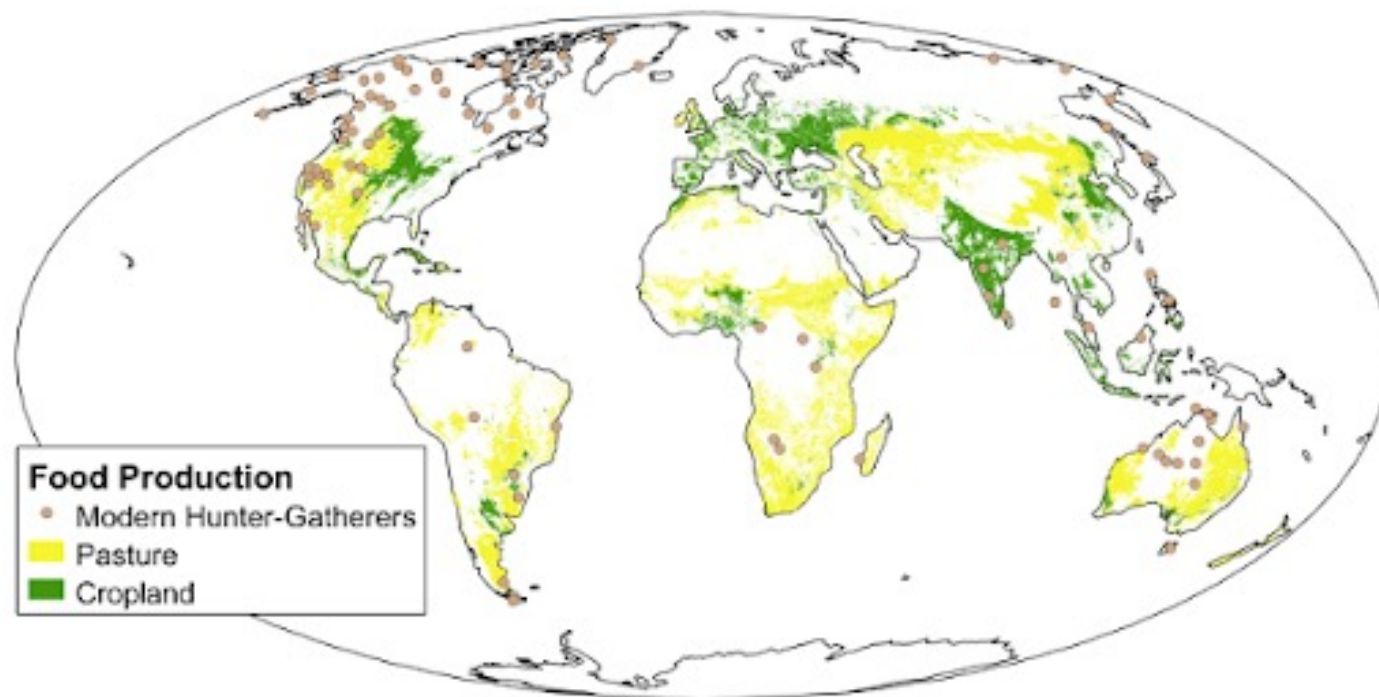
Mismatch between **evolutionary environment** and current environment

Hunter Gatherer lifestyle

- Immediate return
- No cultivation or domestication
- Many important unknowns!



Finding analogs in the modern world



- Immediate return
- No cultivation or domestication
- Many important unknowns!



How good are modern hunter-gatherers as analogs?

‘Simple’ hunter-gatherer societies:

- Low levels of cultivation & domestication; immediate return subsistence
- Multi-family camps with multilocal (bilocal) residence
~30-50 people *Husband can move to wife's family or vice versa, 50/50 but random*
- Mobile residence with fluid camp composition. *~7 moves/year* (Marlowe 2005)
whole camp doesn't have to move together, subsets moving *Camp will move*
- Egalitarian political organisation
- Predominantly serially monogamous with ~10% of women married polygynously (ibid.) *→ Stable pair bonds, if it breaks down you marry someone else*
- Often, sexual division of labour
→ Males spend more time hunting and females spend more time gathering

- Immediate return
- No cultivation or domestication
- Many important unknowns!



How good are modern hunter-gatherers as analogs?

'Simple' hunter-gatherer societies:

- Low levels of cultivation & domestication; immediate return subsistence
not none! Domestic dogs, fructing
- Multi-family camps with multilocal (bilocal) residence
- Mobile residence with fluid camp composition. ~7 moves/year (Marlowe 2005)
- Egalitarian political organisation
- Predominantly serially monogamous with ~10% of women married polygynously (ibid.)
- Often, sexual division of labour

Not 100% immediate return due to proximity w/ non HGS, can go to market

what we can't reconstruct via archaeology



- Immediate return
- No cultivation or domestication
- Many important unknowns!



How good are modern hunter-gatherers as analogs?

It turns out... quite poor!

1 No one group is a 'living fossil'

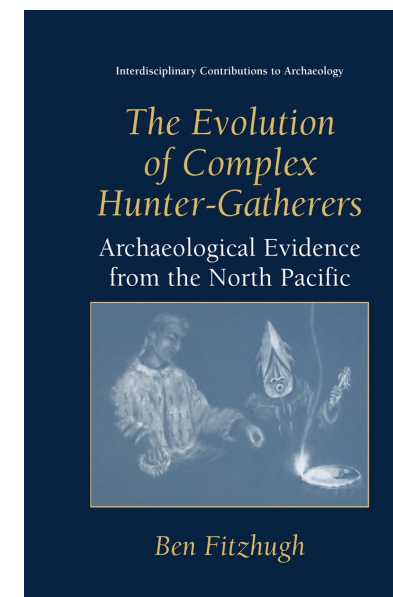
2 Variability

only describes simple HGs

- The HG archetype earlier does not describe **complex, equestrian** or **arctic HGs**.
- Even if we restrict our analyses to *simple* warm-climate HGs vary in:
 - infant mortality (10-46%)
 - fertility rate (0.81-8.5)
 - polygyny (0-90%)
 - sexual division of labour (e.g., net-hunting among Aka)
 - sexual division of calories produced (Ache men provide 87%; Efe men 40%)
- Presumably this variation was **more pronounced in the past**
- so does it even make sense to reconstruct one 'ancestral' lifestyle?

variability was likely the norm

Have different social structures & different diets



How good are modern hunter-gatherers as analogs?



"San Bushmen experience in the Kalahari "

Safaris

reviews

East, Botswana

See all 11 reviews

Bush Ways Safaris is a mobile safari operator. Since its... read more

Other Recent Reviews

ding" 01/11/2019

response" 01/28/2019

1 reviews



Alcoholism in hunter-gatherers → not good proxies for ancestral health

Mismatch-with what?

Mismatch and consequences

Schizophrenia

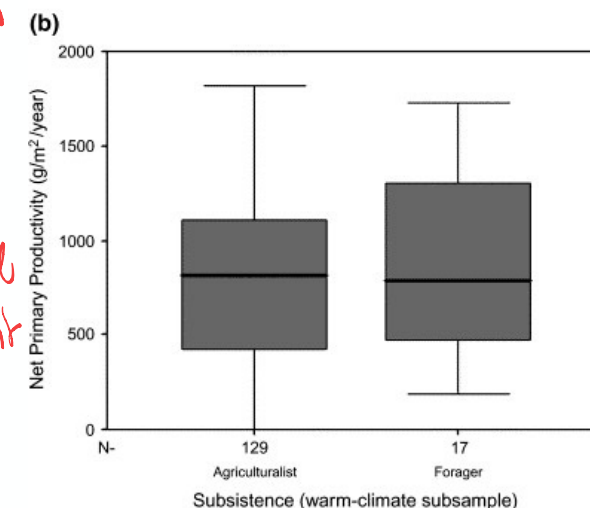
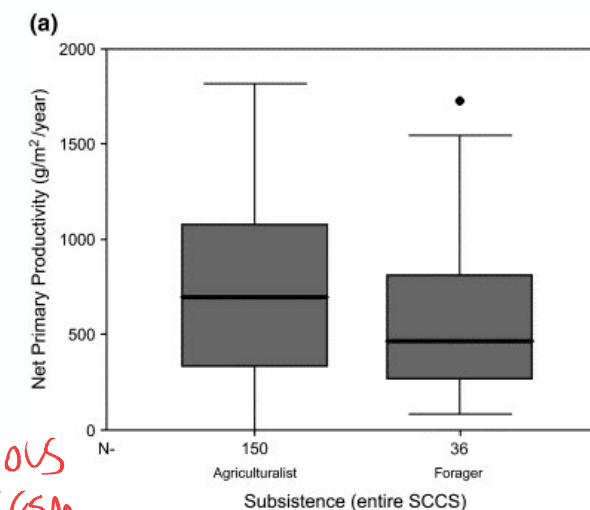
How good are modern hunter-gatherers as analogs?

It turns out... quite poor!

- 1 No one group is a 'living fossil'
- 2 Variability
- 3 No 'pristine' hunter-gatherer societies

- Critics of HG studies state extant HGs have been pushed into **low productivity environments and marginalised** by more successful farming societies.
- Thus extant HGs have much **more nutritional stress and occupy a limited range of habitats compared to our ancestors**, which has knock on effects for all aspects of social organisation (Barker 1999).
- But: When cold-climate foragers are excluded, forager habitats aren't less productive.

These absolutely hold



More + various criticisms from 1 and 2 maybe doesn't hold

How good are modern hunter-gatherers as analogs?

It turns out... quite poor!

- 1 No one group is a 'living fossil'
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arent
very
good
models

Nonetheless, a better model than WEIRD societies.

How do hunter-gatherer societies differ from WEIRD societies, and might these variables explain the high rates of mental health issues in WEIRD societies?



Assuming these rates
are lower in HGs...
but are they?!

rates of
mental health
issues

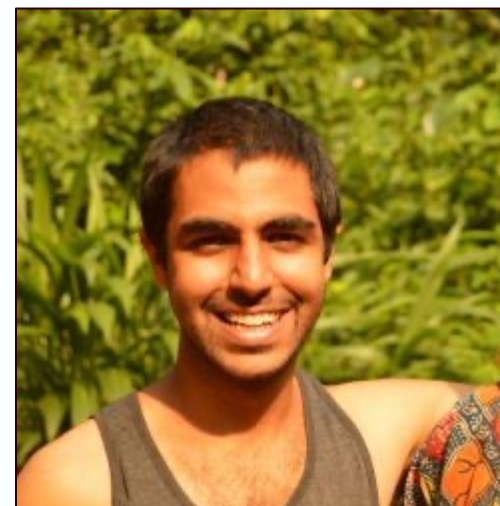
Hunter-gatherer mental health is a shockingly under-researched area

“Psychiatric epidemiological research with hunter-gatherer populations is **limited to a handful of studies**. There are equally few non-medical studies examining subjective well-being and happiness”

We don't
really know
how HG
mental health
differs from
non HGs

No
actual
studies?

↓
Some indication
that HGs
have lower
rates of
mental illness



Dr. Nikhil Chaudhary



Assuming these rates
are lower in HGs...
but are they?!

Key



- 1 No one group is a 'living fossil'
- 2 Variability
- 3 No 'pristine' hunter-gatherer societies
- 4 Not much is known about mental health in simple hunter-gatherer societies

Still a better model than WEIRD societies.

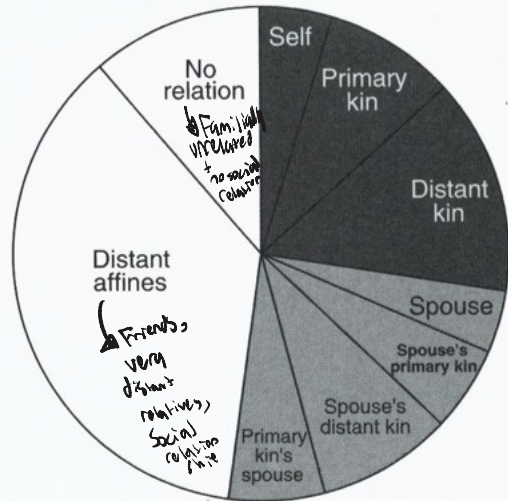
How do hunter-gatherer societies differ from WEIRD societies, and might these variables explain the high rates of mental health issues in WEIRD societies?

Potentially significant causes of mismatch

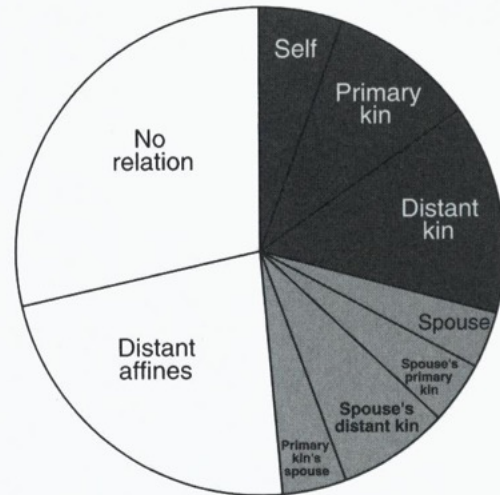
- 1 Group living
- 2 Shared ritual
- 3 (In)equality
- 4 Childcare

Group living seems the norm

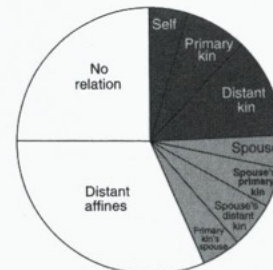
who's in your group as a HG?



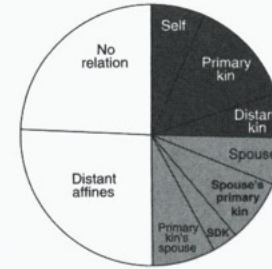
Agta



Mbendjele



Ju/'hoansi



Ache



Neuron, Vol. 35, 395–405, July 18, 2002, Copyright ©2002 by Cell Press

A Neural Basis for Social Cooperation

feels rewarding
to cooperate w/ people

James K. Rilling,^{1,2} David A. Gutman,
Thorsten R. Zeh, Giuseppe Pagnoni,
Gregory S. Berns, and Clinton D. Kilts
Department of Psychiatry and Behavioral Sciences
Emory University
Atlanta, Georgia 30322

expertise, information, oppor-
tunity, resources.

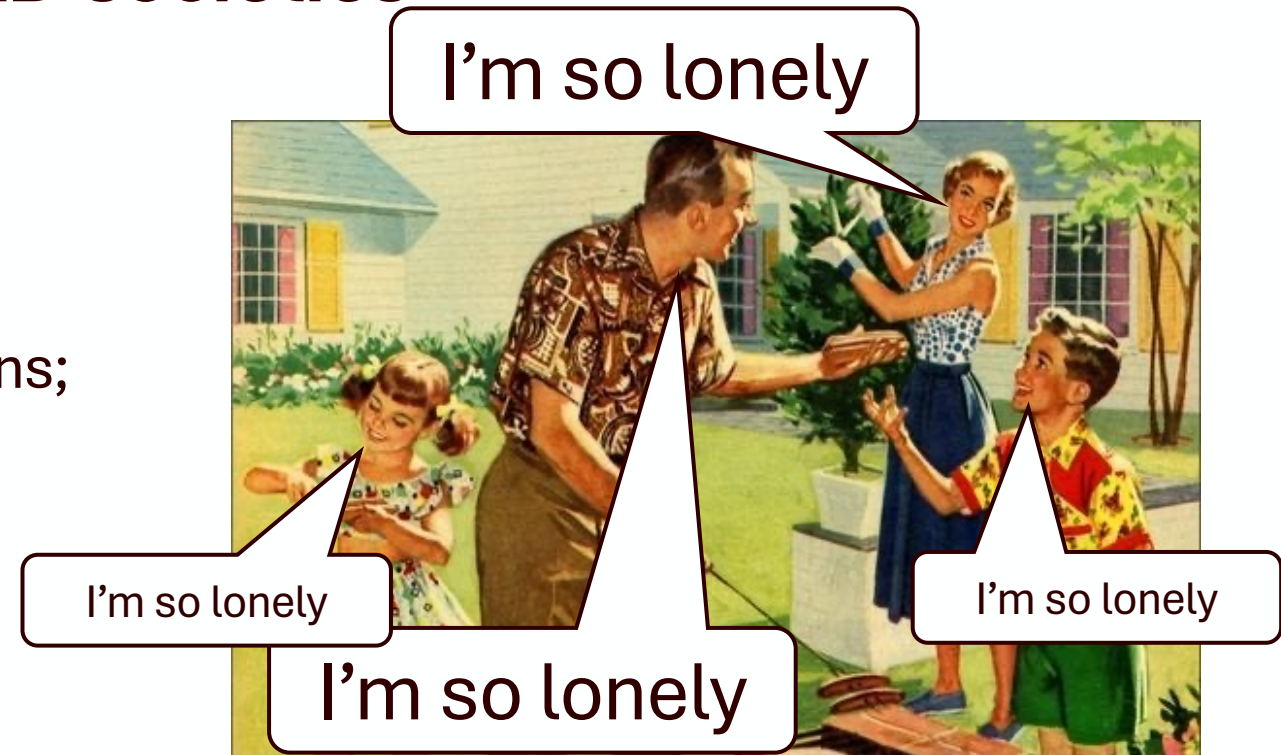
On the other hand, cooper-
ation is rare in the rest of
has only been convincingly d
of species (Axelrod and Han
Trivers, 1971). In attempting

Group living is very rare in WEIRD societies

UK:

- 1-person households as high as 35%
- 25% of adults have no contact with cousins; 80% see them less than once a month
- 35% exchange favours with neighbours

↳ Norm should be sharing food w/ others



Levels of social isolation are very high

Evolved proximate mechanisms to avoid social isolation (anxiety, feeling of loneliness, stress) kick in

Isolation → anxiety, stress etc
mechanisms

Shared initiation and ritual is common in hunter-gatherer life



Why evolve proximate mechanisms to make these feel good?

- Group life & food sharing key to the human foraging niche
- Cultural group selection?

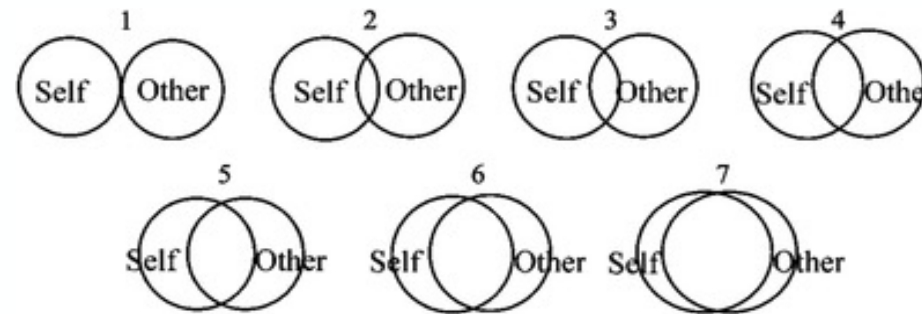
↳ Not certain if rituals feel good
is due to group level selection vs
individual level selection

↳ rituals promote
group cohesion and
membership

Synchronous movement and chanting:

- Oxytocin
- β -endorphins

Increased trust, cooperation, and other-self merging



Shared initiation and ritual is not common in WEIRD life

Suicide risk falls with increasing religious service attendance

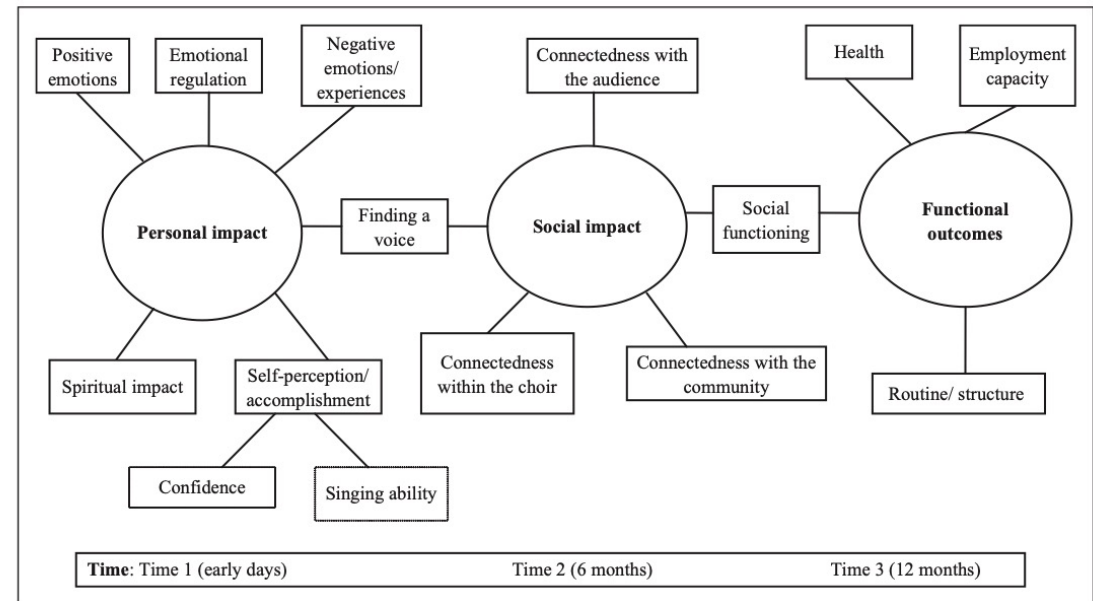
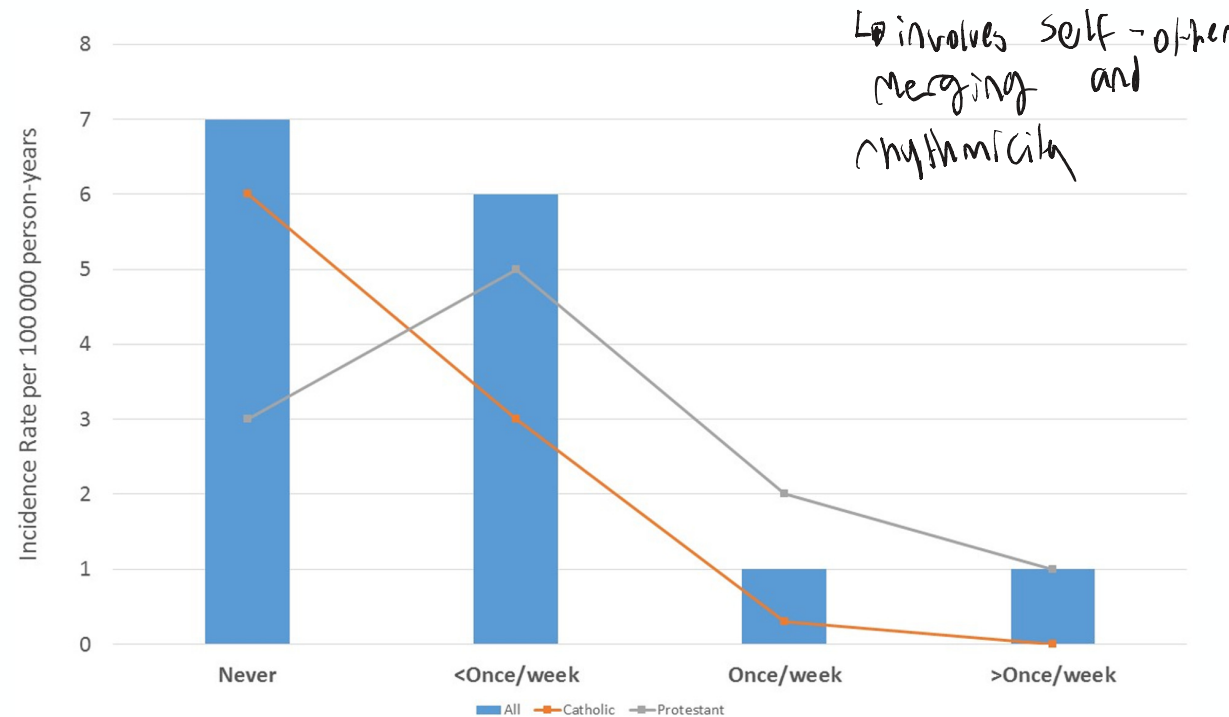


Figure 1. Thematic map derived from interviews with choir participants.

Social and mental health benefits of choir singing in disadvantaged adults

Handwritten note: Therapeutic are important for mental health

Egalitarianism and a lack of inequality is the norm in hunter-gatherer societies

Economic equality:

- No storage → can't hoard → can't accumulate wealth
- Weak concept of ownership
- Sharing

→ You can split off from group if there is a problem leading to...

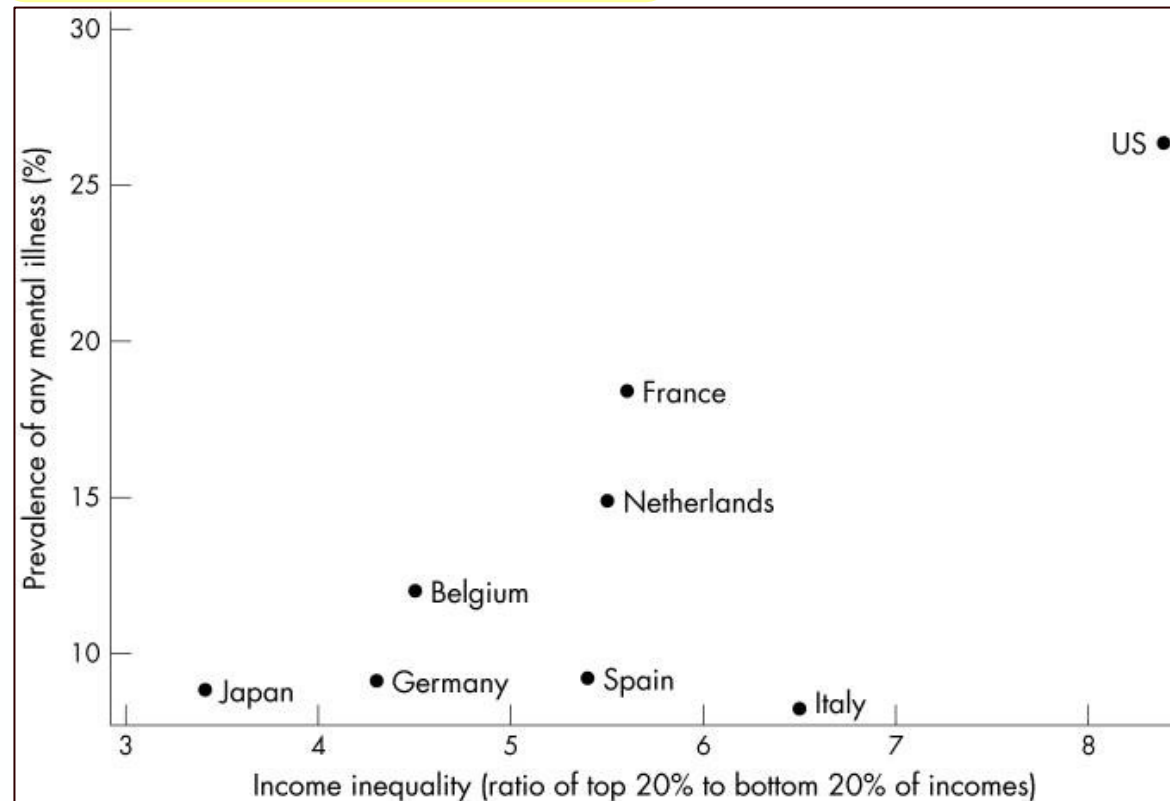
Political/social equality:

- Consensus decision-making
- Leadership ranks absent or weak
- Rejection of dominant behaviour
- Emphasis on autonomy

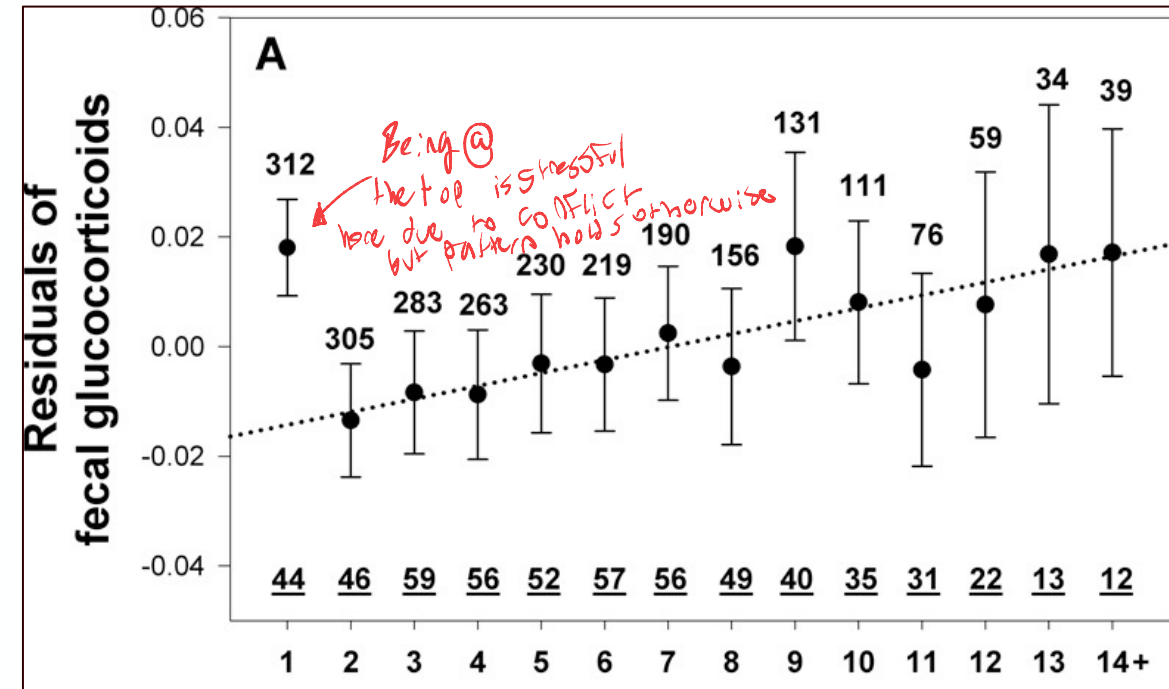


Inequality is the norm in WEIRD societies

Economic inequality correlates positively with prevalence of mental illness



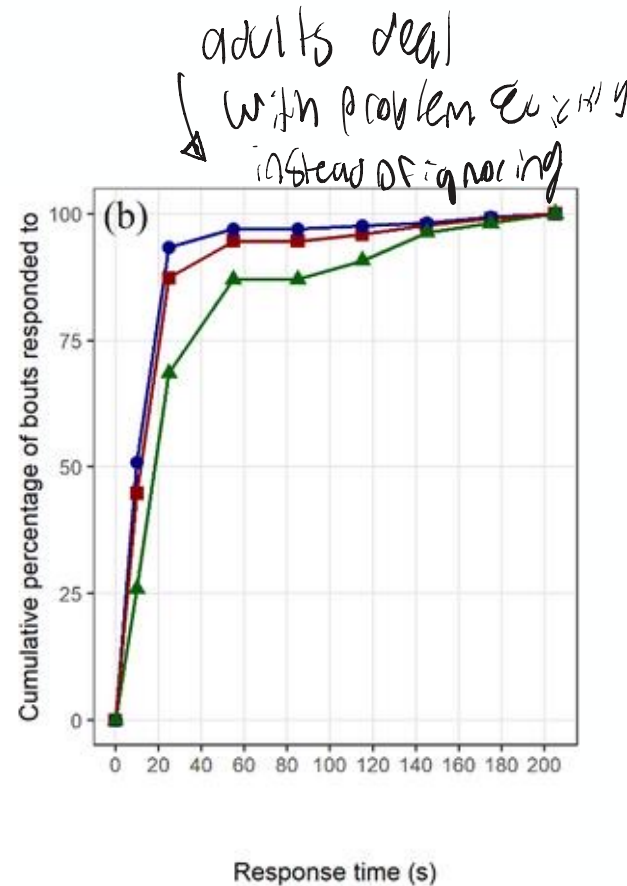
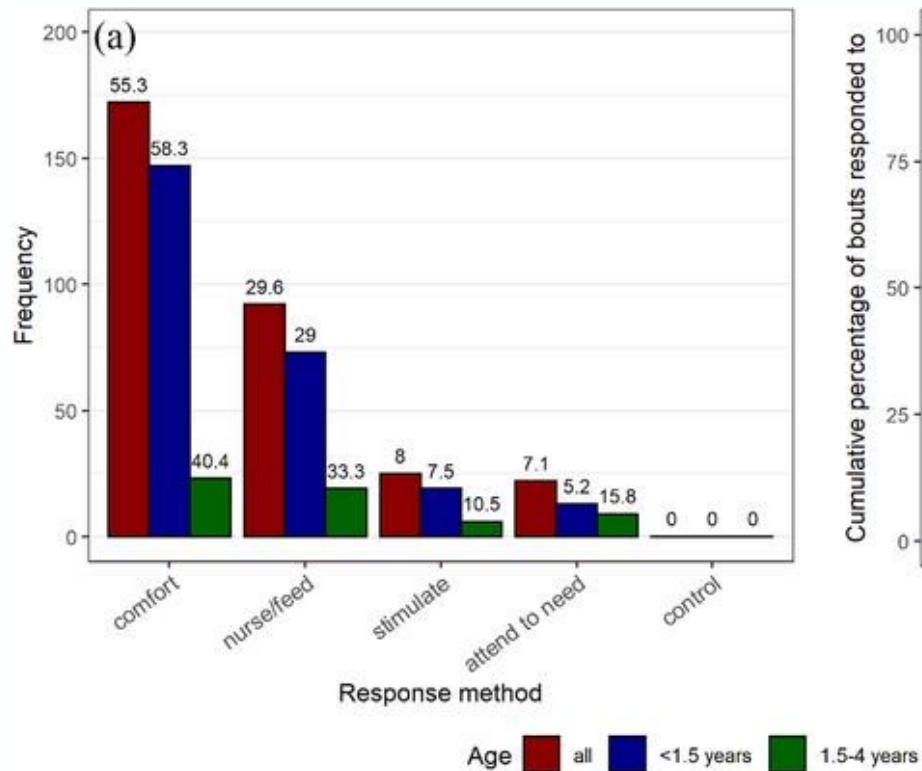
What's the mechanism?



Deep primate mechanisms: hierarchies generate stress

↳ Being at the bottom of a hierarchy increases stress.
Less autonomy, power, resources → more stress

Responsive and soothing care is the norm in hunter-gatherer society



Childcare responsiveness mismatch (?)

The Journal of Child
Psychology and Psychiatry

Journal of Child Psychology and Psychiatry 61:11 (2020), pp 1184–1193



doi:10.1111/jcpp.13223

Leave baby to
cry when crying

Parental use of 'cry it out' in infants: no adverse effects on attachment and behavioural development at 18 months

Ayten Bilgin,¹ and Dieter Wolke^{1,2}

¹Department of Psychology, University of Warwick, Coventry, UK; ²Division of Health Sciences, Warwick Medical School, University of Warwick, Coventry, UK

Debate

The Journal of Child
Psychology and Psychiatry

Journal of Child Psychology and Psychiatry 62:12 (2021), pp 1488–1490



doi:10.1111/jcpp.13390

Commentary: Does 'cry it out' really have no adverse effects on attachment? Reflections on Bilgin and Wolke (2020)

Abi M. B. Davis,  and Robin S. S. Kramer

School of Psychology, University of Lincoln, Lincoln, UK

Closeness and touch is the norm in hunter-gatherer society

Table 3
Closeness, Close Care, and Allomaternal Involvement

Age group (years)	Close care			
	Total (hr)	% Allomother		
<1.5	8.8 (2.8)	41.9 (25.6)		
1.5–4	3.9 (2.2)	42.8 (25.1)		

Held		Physical contact	
Total (hr)	% Allomother	Total (hr)	% Allomother
5.3 (2.2)	39.3 (31.5)	9.2 (2.0)	40.3 (25.8)
1.8 (1.4)	40.3 (36.8)	7.0 (2.3)	45.3 (21.5)



Childcare responsiveness mismatch (?)

Table 1
Mean hours per day of mother–infant skin-to-skin contact for the skin-to-skin and control groups through 4.

Group	Infant age	
	Week 1	Weeks 2–4
Skin-to-skin	5.03 (1.38)	2.77 (1.01)
Control	0.47 (0.90)	0.20 (0.42)

Note: Standard deviations are in parentheses.



Mothers and babies seem to expect high levels of contact:

- Physical contact is associated with secure attachment on the part of the baby, and decreased risk of maternal depression
- Interventions (babywearing; infant massage) shown to correlate with the same!

promoting touch
secure attachment



Key

Potentially significant causes of mismatch

- 1 Group living
↑ evolution selected for mechanisms for feeling good in a group
- 2 Shared ritual
- 3 (In)equality
- 4 Childcare

All can be linked to proximate mechanisms that relate to mood and mental wellbeing

For good evolutionary reasons!

But detailed research is lacking

↳ esp WRT hunter-gatherer societies